Colegio de Montalban

Kasiglahan Village, San Jose, Rodriguez, Rizal

Institute of Computer Studies

WEB- BASED NOTE SHARING, RATING, AND ANNOUNCEMENT SYSTEM.

Leader:

Jallorina, Xyrene Carmelyn A.

Members:

Carabot, Sharine T.

Encarnacion, Austin G.

Raya, John Ruzzel B.

CHAPTER 1

INTRODUCTION

Background of the Study

In the modern academic environment, digital transformation plays an important role in improving how students learn and share information. According to Wang (2025), modern academic environments frequently encounter challenges regarding disorganized learning materials and fragmented file sharing, which underscores the necessity of centralized web platforms to enhance student collaboration and resource management.

In educational settings, platforms such as Google Classroom have been widely adopted due to their accessibility, intuitive interface, and integration capabilities for managing instructional materials. A study by Al-Mhiqani et al. (2023) highlighted that utilizing structured e-learning and management platforms significantly enhances student engagement, simplifies the distribution of learning resources, and maintains academic continuity during remote or flexible learning arrangements. This indicates that consolidating academic interactions and files into a single, organized platform helps bridge communication gaps between students and educators

Students still use everyday messaging apps for school communication. In the Philippines, Pedroso (2023) found that students heavily rely on platforms like Facebook Messenger for academic inquiries and file sharing. This shows that while these apps provide

accessibility for student communication, they lack proper educational management features, often resulting in unorganized requirements and lost announcements.

At Colegio de Montalban (CDM), notes are usually shared through group chats and direct messages, which leads to unorganized files. There is no assurance that the shared files are correct or updated. Also, announcements and academic requirements are often posted across fragmented channels such as being mixed with personal chats on Messenger or scattered inside separate Google Classroom spaces which causes students to miss crucial deadlines. Based on the interviews conducted with a student and a professor in the school, they experience a hard time finding scattered files and missing important updates. Aside from this, since messaging apps handle both personal and school matters, academic inquiries easily get buried under a flood of personal conversations, making it difficult for students to focus and hard for professors to reply to everyone on time.

Even with many digital tools available, there is still no system built specifically for note sharing, ratings, and announcements inside the school. The school needs a system made just for academics, where files are safe, easy to find, and well-managed

To address these problems, this study proposes a Web-Based Note Sharing, Rating, and Announcement System as a Progressive Web App (PWA) named NoteSphere. The system provides a secure account registration and verification process where users can register, review their entered subject codes through a confirmation card, and verify their accounts using an email OTP before logging in.

Once inside, role-based access is implemented for students, professors, and administrators. Students can upload, view, and download notes, with student uploads requiring admin approval to maintain quality. They can also rate the quality of notes using a 1-to-5-star scale and access organized institute-based cards and announcements that may include external links and deadlines. Furthermore, the system includes a Helpboard Q&A forum where users can post questions about notes or other topics that can help them in their studies, with an anonymous posting option. It also provides a dedicated direct messaging feature for consultations, ensuring that student messages remain visible until they are replied to. Additional admin features include School Year Archives, Enrollment Updates, system guides, and customizable Dark and Light mode options.

Statement of the Problem

At present, students and professors at Colegio de Montalban rely on group chats, personal messages, and other informal platforms to share learning materials, post academic announcements, and manage academic tasks. As a result, learning materials may become unorganized, difficult to locate, and outdated. Important announcements may also be overlooked because of information overload on social media and messaging platforms. Direct messages between students and professors may also become buried or unanswered, while the lack of a centralized system makes the monitoring and management of academic files and activities more difficult. These issues may affect students' academic preparation and communication within the institution.

To address these concerns, this study proposes the development of **NoteSphere: A Web-Based Note Sharing, Rating, and Announcement System as a Progressive Web App (PWA)**, with multi-role capabilities for students, professors, and system administrators.

Specifically, this study seeks to evaluate the NoteSphere system based on selected **ISO/IEC 25010 software quality characteristics**, namely Functional Suitability, Performance Efficiency, and Maintainability, through the following questions:

1. What is the level of Functional Suitability of the NoteSphere system in terms of:

a. Functional Completeness, in terms of the extent to which the system provides the required functions, including:

- student self-registration with a review card, editing of registration details, and email OTP verification;
- secure addition and management of professor accounts by the system administrator;
- role-based access for students, professors, and system administrators;
- institute-filtered uploading, downloading, rating, and searching of learning materials;
- professor announcements and LMS tasks filtered according to year level and section;
- student submission and grading functions;
- anonymous Q&A forum;
- faculty-student consultation and messaging; and

  system administrator oversight and approval of student-uploaded learning materials.

b. Functional Correctness, in terms of the extent to which the system accurately processes student registration and OTP verification, manages user accounts and roles, and correctly displays learning materials, announcements, submissions, grades, and messages across the appropriate user dashboards.

c. Functional Appropriateness, in terms of the extent to which the system provides appropriate functions that help students, professors, and system administrators perform their respective academic tasks efficiently.

2. What is the level of Performance Efficiency of the NoteSphere system in terms of:

a. Time Behavior, in terms of the time required to process email OTP verification, search learning materials, upload and download documents, process grading activities, and display announcements and messages.

b. Resource Utilization, in terms of how efficiently the system uses available device and system resources when accessed as a Progressive Web App (PWA) on mobile and desktop devices.

c. Capacity, in terms of the system's ability to accommodate multiple users, pending student registrations, and uploaded learning materials while maintaining acceptable system performance.

3. What is the level of Maintainability of the NoteSphere system in terms of:

a. Modifiability, in terms of how easily the system can be modified or updated to support future features and improvements across the student, professor, and administrator modules.

b. Testability, in terms of how easily the system functions, database operations, and role-based access controls can be tested.

c. Analysability, in terms of how easily system errors, database issues, and other technical problems can be identified and analyzed for resolution.

d. Reusability, in terms of the extent to which system components or modules can be reused or adapted for future academic systems.

Objectives of the Study

The researchers aim to develop **NoteSphere: A Web-Based Note Sharing, Rating, and Announcement System as a Progressive Web App (PWA)** for Colegio de Montalban. The system is intended to provide a centralized platform for students, professors, and system administrators to organize and access learning materials, support secure student registration

through email OTP verification, facilitate academic announcements and LMS tasks, manage student submissions and grades, and improve academic communication.

Specifically, this study aims to:

1. Determine the level of Functional Suitability of the NoteSphere system in terms of:

a. Functional Completeness, by determining whether the system provides the required functions for:

student self-registration with a review card, editing of registration details, and email OTP verification;

secure addition and management of professor accounts by the system administrator;

role-based access for students, professors, and system administrators;

institute-filtered uploading, downloading, rating, and searching of learning materials;

professor announcements and LMS tasks filtered according to year level and section;

student submission and grading functions;

anonymous Q&A forum;

faculty-student consultation and messaging; and

system administrator oversight and approval of student-uploaded learning materials.

b. Functional Correctness, by determining whether the system accurately processes student registration and OTP verification, manages user accounts and roles, and correctly displays learning materials, announcements, submissions, grades, and messages across the appropriate user dashboards.

c. Functional Appropriateness, by determining whether the system provides appropriate functions that help students, professors, and system administrators perform their respective academic tasks efficiently.

2. Measure the level of Performance Efficiency of the NoteSphere system in terms of:

a. Time Behavior, by measuring the time required to process email OTP verification, search learning materials, upload and download documents, process grading activities, and display announcements and messages.

b. Resource Utilization, by assessing how efficiently the system uses available device and system resources when accessed as a Progressive Web App (PWA) on mobile and desktop devices.

c. Capacity, by determining the system's ability to accommodate multiple users, pending student registrations, and uploaded learning materials while maintaining acceptable system performance.

3. Assess the level of Maintainability of the NoteSphere system in terms of:

a. Modifiability, by assessing how easily the system can be modified or updated to support future features and improvements across the student, professor, and administrator modules.

b. Testability, by assessing how easily the system functions, database operations, and role-based access controls can be tested.

c. Analysability, by assessing how easily system errors, database issues, and other technical problems can be identified and analyzed for resolution.

d. Reusability, by assessing the extent to which system components or modules can be reused or adapted for future academic systems.

Significance of the Study

This study on the development of NoteSphere: A Web-Based Note Sharing, Rating, and Announcement System (PWA) will provide significant benefits to the following stakeholders:

- Administrators of Colegio de Montalban. This study will help school leaders and administrators by providing a centralized and secure platform that improves

institutional communication, ensures the organized distribution of academic announcements, and fosters a collaborative digital environment within the campus.

- Instructors and Professors. The system will benefit teachers by giving them a reliable channel to post announcements with external learning links (such as Google Forms, Meet, and YouTube). It will also help them manage student consultations properly through direct messaging that supports files, pictures, and voice records, preventing important academic queries from getting lost.
- Students. As the primary users, students will gain direct access to a structured and reliable platform where they can easily upload, search, and download study notes. The rating feature will help them identify high-quality materials, while the anonymous Q&A forum will give them a safe space to ask academic questions without hesitation. Furthermore, the secure OTP registration and PWA accessibility ensure a lightweight and safe user experience.
- Future Researchers. This study will serve as a valuable reference, guide, or baseline data for future students and researchers who plan to develop, enhance, or conduct similar studies on academic web applications, Progressive Web Apps (PWAs), and digital collaboration systems.

Scope and Delimitation

This study focuses on building and testing NoteSphere: A Web-Based Note Sharing, Rating, and Announcement System as a Progressive Web App (PWA) for Colegio de Montalban (CDM).

1. Users: The system is built for three types of users: Students, Professors, and Admins.

2. Main Features:

Secure registration where users can check their details using a review card and verify their accounts through email OTP.

- Note-sharing where users can upload, search, download, and rate reviewers from 1 to 5 stars.
- Announcements from professors with links (like Google Forms, Meet, or YouTube) and deadlines.
- A Helpboard Q&A forum where users can post questions openly or anonymously.

Direct messaging for student-teacher consultations that can send files, pictures, and voice records.

- Admin features like archiving school years, updating enrollments, system guides, and Dark/Light mode options.

3. Evaluation: The system will be tested and graded by selected students, teachers, and admins using the ISO/IEC 25010 standards, focusing on Functional Suitability, Performance Efficiency, and Maintainability for the school year 2026–2027.

Delimitation:

1. What it cannot do: The system is only for sharing notes, announcements, Q&A, and messaging. It does not cover enrollment processing, official grading systems, or school payments.

2. Who can use it: It is only for the students, teachers, and admins of Colegio de Montalban.
3. Technical limits: Because it is a web app, users need an internet connection to use it; it cannot work fully offline. Also, the speed and performance can depend on the user's internet connection and device.

Operational Framework



Figure 1

An Operational Framework illustrating the development of the Web-Based Note Sharing, Rating, and Announcement System

The project developers utilized the **Agile Methodology** to guide the development of NoteSphere as a Progressive Web App (PWA). The Agile process provides an iterative and

organized approach to system development through activities such as **Requirement Gathering, System Design, Development, Testing, Deployment, and Review**. These activities allow the developers to continuously assess the system, identify necessary improvements, and make adjustments based on requirements and evaluation results.

The process begins with **Requirement Gathering**, where the developers identify the needs of students, professors, and system administrators. This includes requirements for student registration and email OTP verification, learning material management, note rating, announcements, LMS tasks, student submissions and grading, Q&A, consultation and messaging, and administrative management. The identified requirements serve as the basis for defining the functions and features of the system.

The next stage is **System Design**, where the system architecture, database structure, user interfaces, navigation, and role-based functions are planned based on the identified requirements. This is followed by **Development**, where the actual coding and integration of the system components are performed to create the NoteSphere web application.

After development, the system undergoes **Testing** to identify functional errors, database issues, interface problems, and other system defects. Necessary corrections and improvements are then implemented. The system proceeds to **Deployment**, where the developed NoteSphere system is made available for its intended users. The **Review** stage involves gathering user feedback and evaluating the system based on the selected ISO/IEC 25010 software quality characteristics, specifically **Functional Suitability, Performance Efficiency, and Maintainability**. The results of the review and evaluation are used to identify areas for further improvement and refinement of the system.

The developed NoteSphere system focuses on providing a centralized academic platform for students, professors, and system administrators at **Colegio de Montalban**. The system includes secure student registration with a review card and email OTP verification; learning material uploading, downloading, rating, and searching; professor announcements and LMS tasks with year-level and section filtering; student submission and grading functions; an anonymous Q&A forum; faculty-student consultation and messaging; and administrative monitoring and approval of student-uploaded learning materials.

The system will be evaluated by selected students, professors, and system administrators who will use the system and answer the evaluation questionnaire. The evaluation will focus on **Functional Suitability, Performance Efficiency, and Maintainability** based on the selected ISO/IEC 25010 software quality characteristics during the academic year **2026–2027**.

The study is limited to the academic functions provided by NoteSphere, particularly learning material sharing, note rating, academic announcements, LMS tasks, student submissions and grading, Q&A, consultation and messaging, registration, and administrative management. It does not cover other institutional processes such as online enrollment, financial management, payroll, or other administrative systems outside the defined scope of the project.

Furthermore, NoteSphere is intended for authorized users within Colegio de Montalban and require s an internet connection for its web-based and PWA functions. The system's performance may also be affected by factors beyond the researchers' control, such as internet connection speed, device capabilities, browser compatibility, server resources, and the users' level of technical familiarity.

Conceptual Framework

The researchers used the Input-Process-Output (IPO) model to show how the study works:

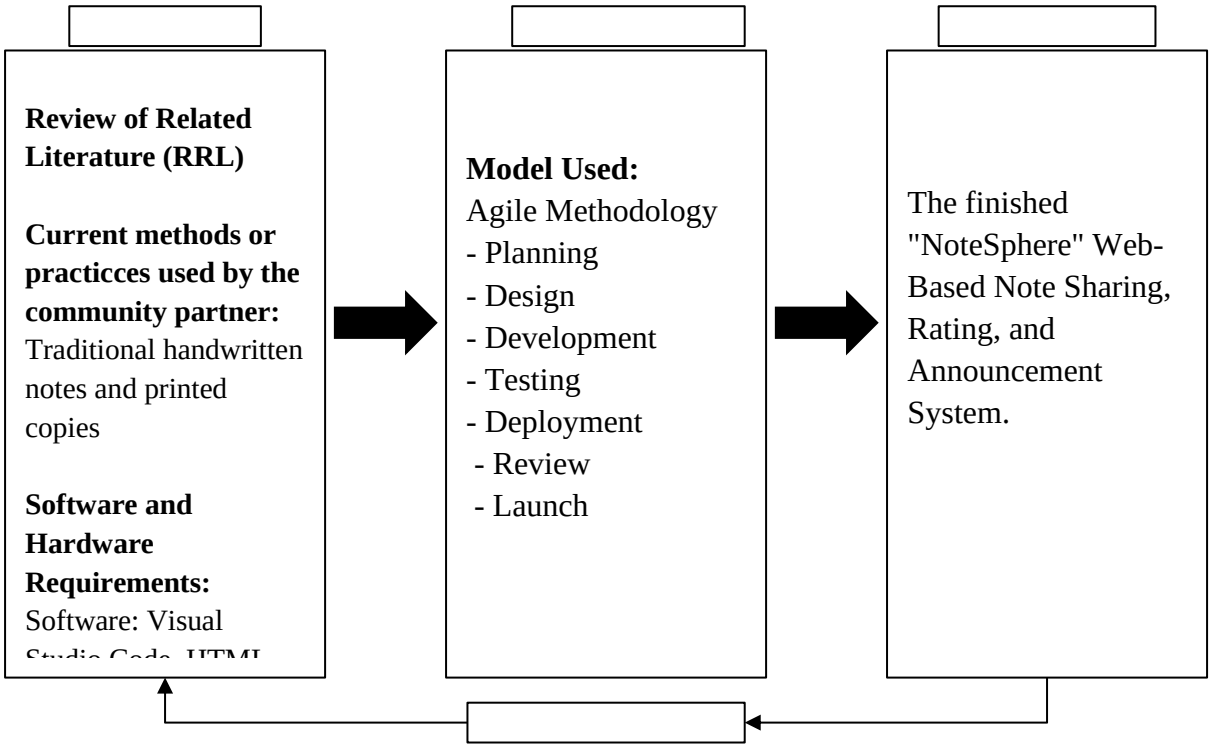


Figure 2

A Conceptual Framework illustrating the development of the Web-Based Note Sharing

The conceptual framework's **Input Phase** contains all the essential requirements used in developing the NoteSphere system. The inputs include the Review of Related Literature (RRL), existing informal sharing methods used at Colegio de Montalban, user requirements, and the identified needs of students, professors, and system administrators. These include secure student registration with email OTP verification and a review card, learning material sharing, note ratings, announcements with external links, anonymous Q&A, direct messaging with support for files, pictures, and voice records, and other system requirements. The inputs

also include the software and hardware requirements such as Visual Studio Code, HTML, CSS, JavaScript, PHP, MySQL, computers, and an internet connection. These inputs serve as the foundation for the development of NoteSphere.

In the **Process Phase**, the researchers utilized the **Agile Methodology**, which involves requirements gathering, system design, coding and development, testing, deployment, and review and evaluation. During the evaluation phase, the developed system is assessed using selected characteristics of the **ISO/IEC 25010 software quality model**, specifically **Functional Suitability, Performance Efficiency, and Maintainability**. The results of the evaluation may be used to identify areas for improvement and further system refinement.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Foreign Literature

Salimi et al. (2025) investigated the relationship between students' digital literacy, knowledge sharing, and academic performance in online learning environments. The researchers surveyed 330 graduate students from a public university in Iran and used structural equation modeling to examine the relationships among the variables. The findings showed that digital literacy positively and significantly predicted knowledge sharing and academic performance, while knowledge sharing also mediated the relationship between digital literacy and academic performance. This study is relevant to the present study because NoteSphere promotes digital knowledge sharing by providing students and professors with an organized platform for sharing, accessing, and managing academic notes and learning materials.

Bhargava et al. (2022) examined digital platform ecosystems and how platform design influences the participation of different groups of users. The study discussed how digital platforms connect content producers and users and how platform design can influence the supply and distribution of digital content. The findings emphasized the importance of platform design in supporting participation and maintaining an effective digital ecosystem. This study is relevant to the present study because NoteSphere is designed as a digital academic platform that connects students, professors, and administrators while allowing authorized users to contribute to and access educational content.

Talele et al. (2025) examined the development of a multi-thread file-sharing system using Java Sockets and multithreading. The study focused on improving the process of sharing files between users through a network-based system and addressed technical considerations involved in handling multiple file-sharing operations. The study demonstrates the importance of efficient digital file-sharing mechanisms in systems that allow users to exchange digital resources. This is relevant to the present study because NoteSphere includes file management and sharing functions that allow students and professors to upload, access, and share academic notes and learning materials.

Li and Zhu (2022) examined the factors affecting college students' satisfaction and acceptance of online learning platforms in online and blended learning environments. The study focused on system quality, information quality, and user satisfaction. The findings showed that information quality had an important effect on students' satisfaction with online learning platforms, particularly in fully online learning environments where students rely more heavily on digital resources. This study is relevant to the present study because NoteSphere aims to provide students with an organized online platform where academic notes and learning materials can be accessed, managed, and evaluated based on their usefulness.

Wu et al. (2025) examined the relationship between the quality of digital learning resources and college students' learning engagement. The researchers collected data from 606 college students and used structural equation modeling to examine how digital learning resource quality affects students' engagement. The findings showed that the quality of digital learning resources can directly and indirectly influence learning engagement through attention control and academic self-efficacy. This study is relevant to the present study because

NoteSphere provides digital learning resources through shared academic notes, while its rating feature can help students identify useful and quality learning materials.

Cheung et al. (2022) examined university students' perceptions of the usefulness of open educational resources when learning shifted from face-to-face to online environments. The study gathered responses from students in different learning modes and found that open educational resources became more useful during online learning, particularly for accessing additional course materials and preparing for academic requirements. However, students also expressed concerns regarding the accuracy and completeness of online resources. This study is relevant to the present study because NoteSphere provides students with centralized access to academic learning materials while its admin approval and rating features can help promote more reliable and useful shared notes.

Kelb et al. (2024) examined the types of online resources used by university science and mathematics students to supplement or replace classroom lectures. The study found that more than 80% of the students used online resources such as YouTube and Open Educational Resources to improve their understanding of course content and access additional learning materials. The researchers also emphasized the importance of students' ability to search for and critically evaluate the quality of online resources. This study is relevant to the present study because NoteSphere provides an organized academic platform where students can access additional notes and learning materials and use ratings to help identify useful resources.

Wang et al. (2025) examined the impact of digital resource utilization on student performance in distance learning. The study surveyed students from Allama Iqbal Open University and analyzed 308 valid responses using the DeLone and McLean Information

Systems Success Model. The findings showed that knowledge quality, digital resource system quality, and service quality positively affected students' performance in the distance learning environment. This study is relevant to the present study because NoteSphere provides a digital platform for accessing and managing academic learning materials, making the quality and organization of digital resources important considerations in the proposed system.

Colbert et al. (2016) developed instruments for evaluating the quality of online educational resources using ratings from trainees. The researchers reduced a list of quality indicators into rating items and asked trainees to evaluate 39 online educational resources using Likert-scale measures. The study developed two scoring instruments, the METRIQ-8 and METRIQ-5, which showed relationships with faculty evaluations and external certification. This study is relevant to the present study because NoteSphere uses a rating feature that allows students to evaluate shared academic notes and helps other users identify learning materials that may be useful and of good quality.

Sims (2025) examined the use of a learning management system for online education and found that the announcement tool served as a primary means of communicating with remote students. The study observed that announcements were used for course instructions, clarifications, lecture links, cancellations, assessment schedules, submissions, marks, feedback, and course resources. The findings emphasized the importance of clear and coordinated communication when using announcements in an online learning environment. This study is relevant to the present study because NoteSphere includes an announcement feature that allows professors and administrators to provide students with academic updates, deadlines, links, and other important information in a centralized platform.

Bina and Anggraini (2025) evaluated the quality of a web-based e-learning platform using the ISO/IEC 25010 software quality model and an end-user questionnaire. The study involved students and instructors and assessed functional suitability, usability, reliability, and performance efficiency. The findings showed that the evaluated platform achieved an overall quality rating categorized as good, although performance efficiency required further improvement. This study is relevant to the present study because NoteSphere will also be evaluated as a web-based educational system using selected ISO/IEC 25010 software quality characteristics, particularly Functional Suitability and Performance Efficiency, to determine whether its functions and performance meet the requirements of its intended users.

Aziz and Irfan (2026) evaluated a web-based internship monitoring information system for vocational education using the ISO/IEC 25010 software quality model. The study assessed functional suitability, usability, reliability, and performance efficiency. The results showed that the system achieved 100% functional suitability and 96.57% reliability, while the usability results indicated a positive user experience and the performance was considered acceptable for a web-based application. This study is relevant to the present study because NoteSphere is also a web-based educational system that requires software quality evaluation, particularly in terms of Functional Suitability and Performance Efficiency.

Local Literature

Mangulad (2026) developed a Student Information Management System for Medical Colleges of Northern Philippines and the International School of Asia and the Pacific. The system was designed to centralize student information and automate processes involving

enrollment, grades, schedules, student profiles, and document requests. The system was developed using HTML, CSS, JavaScript, PHP, MySQL, and phpMyAdmin and was evaluated using the ISO/IEC 25010 software quality model. The findings showed an overall weighted mean of 4.62, interpreted as Strongly Agree. This study is relevant to the present study because NoteSphere is also a web-based educational system designed to centralize academic information and services for students and professors, while its software quality will also be evaluated using software quality characteristics.

Grepon et al. (2025) developed School-IntegrIS, an integrated school management system for a state college in Northern Mindanao, Philippines. The study aimed to integrate independent school systems and modules into a single platform to provide easier access and user control. The system was evaluated using the ISO/IEC 25010 software quality model, particularly in terms of functionality, efficiency, and usability. The findings showed an overall evaluation of 4.58, which was interpreted as excellent This study is relevant to the present study because NoteSphere also integrates several academic functions, including learning material sharing, announcements, ratings, and communication, into a single platform for students, professors, and administrators. It also supports the present study's use of selected ISO/IEC 25010 characteristics, particularly Functional Suitability, Performance Efficiency, and Maintainability, in evaluating the developed system.

Jacobe and Pascua (2025) developed a web-based Applicant Tracking and Management System with Decision Support for the Expanded Tertiary Education Equivalency and Accreditation Program at St. Paul University Philippines. The study addressed problems in the existing manual process of storing applicant information and generating reports. The system was developed using the Agile Scrum methodology and evaluated using ISO/IEC 25010. The

findings showed high compliance across software quality characteristics, including functional suitability, performance efficiency, usability, and security. This study is relevant to the present study because NoteSphere is also intended to replace scattered or manual academic processes with a centralized web-based system that provides organized access to information and services.

Tabaog (2025) developed a web-based Alumni Tracer System for the University of Cagayan Valley. The study addressed problems associated with manual tracking and outdated record-keeping practices by providing a system for systematically collecting and managing alumni information. The system was developed for a higher education institution and included ISO 25010 as part of its software quality evaluation. This study is relevant to the present study because NoteSphere also aims to replace scattered academic information with a centralized web-based platform that allows users to access and manage academic resources more efficiently.

Daluyon and Bilog (2025) developed a web-based Customer Satisfaction Survey System at AMA University in Quezon City, Philippines. The study aimed to address delayed reporting and scattered feedback data across campuses. The system was evaluated by 357 respondents, including IT experts, customer satisfaction heads, and customers, using an adapted ISO 25010 software quality checklist. The findings showed that the system was rated Highly Acceptable across functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, with an overall weighted mean of 4.67 This study is relevant to the present study because NoteSphere will also use selected ISO/IEC 25010 software quality characteristics, specifically Functional Suitability,

Performance Efficiency, and Maintainability, to determine whether its functions and services meet the needs of its intended users.

` Concepcion et al. (2026) developed a web-based Smart Queuing and Appointment Management System for Bulacan State University–Bustos Campus. The system was developed using HTML, CSS, JavaScript, PHP, and MySQL and was designed to improve administrative service delivery through online scheduling, automated queue management, and real-time monitoring. The researchers evaluated the system using the Technology Acceptance Model and ISO/IEC 25010:2023, involving students, administrative staff, and IT specialists. The findings showed a general weighted mean of 4.59, indicating high functionality, reliability, and usability. This study is relevant to the present study because NoteSphere is also a PHP and MySQL-based web system intended for an educational institution and requires evaluation of its Functional Suitability and Performance Efficiency.

Lagman et al. (2025) developed and evaluated an enterprise-level information system for digital governance in Philippine State Universities and Colleges. The system focused on integrating research, instruction, and extension activities and was developed using the Agile software development methodology. The researchers evaluated the system using the ISO/IEC 25010 software quality model, covering functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The system obtained an overall weighted mean of 3.73, interpreted as Excellent by IT experts. This study is relevant to the present study because NoteSphere also involves multiple user roles and

academic functions and requires a maintainable system that can be modified, analyzed, tested, and reused as future system improvements are introduced.

Synthesis of the Review

Centralized Digital Learning Resources

The reviewed literature shows that digital learning resources have become important in supporting students' access to academic information and additional learning materials. Salimi et al. (2025) emphasized the role of digital literacy and knowledge sharing in online learning, while Li and Zhu (2022) highlighted the importance of information quality in online learning platforms. Wu et al. (2025), Cheung et al. (2022), Kelb et al. (2024), and Wang et al. (2025) also showed that the quality, usefulness, and organization of digital learning resources can affect students' learning experiences and engagement. These studies support the need for an organized platform where academic materials can be easily accessed and managed. In relation to the present study, NoteSphere provides a centralized platform where students and professors can upload, access, search, and manage academic notes and learning materials.

Academic Communication and Announcements

The reviewed studies also emphasized the importance of organized digital communication in educational environments. Sims (2025) found that announcement tools in learning management systems are commonly used to provide instructions, updates, links, schedules, feedback, and other course-related information. Local studies such as Mangulad (2026), Tabaog (2025), and Jacobe and Pascua (2025) also demonstrated how web-based systems can improve the organization and accessibility of academic information. These findings support the inclusion of an announcement function in NoteSphere, where professors and administrators can provide students with academic updates, deadlines, links, and other important information through a centralized platform.

Learning Material Quality and Rating

Several studies emphasized the importance of evaluating the quality and usefulness of digital learning materials. Cheung et al. (2022) noted that students may have concerns about the accuracy and completeness of online educational resources, while Kelb et al. (2024) emphasized the need for students to search for and critically evaluate digital resources. Colbert et al. (2016) further demonstrated how rating instruments can be used to evaluate online educational resources. Wu et al. (2025) and Wang et al. (2025) also highlighted the importance of digital resource quality in supporting learning. These findings support the rating feature of NoteSphere, which allows users to evaluate shared academic notes and helps students identify learning materials that may be useful. The approval process for student-uploaded materials also supports the organization and monitoring of shared academic resources.

Integrated and Multi-Role Educational Platforms

The reviewed local studies demonstrate the value of integrating different academic functions into a centralized web-based system. Grepon et al. (2025) developed an integrated school management system that combined different school modules into one platform, while Concepcion et al. (2026) developed a web-based system for improving administrative services in an educational institution. Lagman et al. (2025) also demonstrated how an enterprise-level system can integrate multiple activities and support different user roles. These studies show that centralized systems can improve the organization and accessibility of information and services. Similarly, NoteSphere integrates learning material sharing, note ratings, announcements, student submissions, communication, and administrative management into one platform for students, professors, and system administrators.

Research Gap

The reviewed literature and studies show that various web-based and digital educational systems have been developed to support learning resources, academic communication, information management, and software quality evaluation. However, many of the reviewed studies focus on specific functions or individual areas of academic management rather than combining several academic resource-sharing and communication functions in one platform.

First, studies such as Salimi et al. (2025), Li and Zhu (2022), Wu et al. (2025), Cheung et al. (2022), Kelb et al. (2024), and Wang et al. (2025) focused mainly on digital learning resources, information quality, knowledge sharing, and student learning experiences. These studies demonstrate the importance of accessible and quality digital resources, but they do not focus on an institution-specific platform that combines academic material sharing with approval and rating features.

Second, Sims (2025) emphasized the use of announcements for academic communication, while other studies focused on centralized student information, school management, or administrative services. Although these systems support organized communication and information management, the reviewed studies do not appear to combine learning material sharing, note rating, academic announcements, and student-professor communication in one platform designed specifically for the academic needs of the target users.

Third, several studies, including Bina and Anggraini (2025), Aziz and Irfan (2026), Daluyon and Bilog (2025), Concepcion et al. (2026), and Lagman et al. (2025), demonstrated the use of ISO/IEC 25010 in evaluating web-based systems. However, these studies evaluated

different systems and combinations of software quality characteristics. The present study addresses this area by evaluating NoteSphere using the selected ISO/IEC 25010 characteristics of Functional Suitability, Performance Efficiency, and Maintainability.

Therefore, the reviewed literature indicates a need for a centralized academic platform that integrates learning material sharing, student-upload approval, note rating, academic announcements, communication, and administrative management within one system. NoteSphere addresses this gap by providing a Progressive Web App designed for students, professors, and system administrators of the Institute of Computing Studies. The system aims to organize academic resources and communication while providing selected software quality evaluation through Functional Suitability, Performance Efficiency, and Maintainability.

CHAPTER 3

RESEARCH METHODOLOGY

Research Design

The study will employ a mixed-methods approach, combining qualitative and quantitative techniques. Qualitative data will be gathered through structured interviews with a student and a professor to understand their experiences, current challenges, and expectations

regarding traditional academic file sharing and announcements. Quantitative data will be obtained through an evaluation questionnaire based on selected ISO/IEC 25010 software quality characteristics. Designated respondents will provide numerical ratings for Functional Suitability, Performance Efficiency, and Maintainability. The combination of qualitative and quantitative data will allow the researchers to identify user requirements and evaluate the developed system using measurable results.

This research design will help the researchers identify common challenges in traditional academic file sharing and communication, determine the system requirements, and evaluate the quality of NoteSphere. The qualitative findings will be used to support the development and refinement of the system, while the quantitative results will be used to determine the level of Functional Suitability, Performance Efficiency, and Maintainability of NoteSphere: A Web-Based Note Sharing, Rating, and Announcement System as a Progressive Web App (PWA) at Colegio de Montalban.

System Design

The system will be developed using the following design tools:

1. System Flowchart – will be used to illustrate the sequence of system operations, including user registration with email OTP verification, note uploading and downloading, announcement posting, student submissions, grading, and consultation or messaging functions.

2. Entity Relationship Diagram (ERD) – will be used to identify the database entities, including users, announcements, notes, submissions, ratings, and messages, and to establish the relationships among these entities within the system.

3. Data Flow Diagram (DFD) – will be used to illustrate how data moves across different system processes, including student note submission, administrator approval, note access and rating, announcement management, and communication between users.

These design tools will help the researchers organize the system structure, database relationships, and flow of information before and during development. The gathered requirements and user feedback will also be considered in designing the system functions.

Project Development

The project will adopt an iterative development approach guided by Agile methodology. The development process will include requirements gathering, system design, development, testing, deployment, and review and evaluation. The requirements gathered from the preliminary interviews with a student and a professor will be used to identify the necessary system functions and improvements.

During development, the researchers will implement and refine the modules of NoteSphere based on the identified requirements. Testing will be conducted to identify errors and necessary adjustments before deployment. After deployment, the developed system will be evaluated by the designated respondents using the selected ISO/IEC 25010 characteristics of Functional Suitability, Performance Efficiency, and Maintainability. The results of the evaluation will be used to identify areas that may require further improvement.

Data Gathering Procedure

The data gathering process will follow a structured, step-by-step approach. First, formal permission will be obtained from the Colegio de Montalban management to conduct the interviews and system evaluation.

Second, preliminary qualitative data will be gathered through structured interviews with a student and a professor. The interviews will focus on their experiences, challenges, and expectations regarding academic note sharing, learning materials, announcements, and communication.

Third, the researchers will develop and refine the NoteSphere system based on the identified requirements and gathered information.

Fourth, after the system has been developed and prepared for evaluation, the selected respondents will be provided with access to the system and an evaluation questionnaire based on the selected ISO/IEC 25010 characteristics.

Finally, the completed questionnaires will be collected, organized, summarized, and analyzed using the appropriate statistical treatment. The results will be used to determine the level of Functional Suitability, Performance Efficiency, and Maintainability of the developed system and to identify areas for possible improvement.

Research Instruments

The primary research instruments will consist of a structured interview guide and an evaluation questionnaire.

The structured interview guide will be used during the preliminary discussions with a student and a professor. The interview questions will focus on the current challenges encountered in academic file sharing, learning material access, announcements, and communication. The information gathered from the interviews will help the researchers identify user requirements and determine the necessary functions of NoteSphere.

In addition, an evaluation questionnaire based on the selected characteristics of the ISO/IEC 25010 Software Quality Model will be used to evaluate NoteSphere. The questionnaire will measure **Functional Suitability, Performance Efficiency, and Maintainability**. The quantitative responses will provide numerical ratings that will be used to determine the level of software quality of the developed system.

Research Sampling

The study will employ purposive sampling for the selection of participants and respondents. The preliminary interview participants will consist of a student and a professor who have experience with academic note sharing, learning materials, announcements, and communication within the institution.

For the system evaluation, selected students, professors, and system administrators will be invited to evaluate the developed NoteSphere system based on their roles and interaction with the system. The respondents will be selected based on their relevance to the system and their ability to provide informed feedback regarding its functions and performance.

The final number of respondents and their distribution will be determined before the actual system evaluation. This approach will ensure that the evaluation includes users who can provide relevant feedback regarding the developed system.

Research Analysis

The study will employ both qualitative and quantitative analysis.

For the qualitative data, the responses gathered from the preliminary interviews with the student and professor will be organized, summarized, and grouped according to recurring themes. These themes may include difficulties in sharing academic files, locating learning materials, receiving announcements, and communicating with other users. The identified themes will be used to support the determination of system requirements and the development of NoteSphere.

For the quantitative data, the responses from the ISO/IEC 25010-based evaluation questionnaire will be organized and summarized using descriptive statistics. The researchers will determine the weighted mean for each evaluation criterion and its corresponding indicators. The results will be interpreted according to the adopted rating scale to determine the level of **Functional Suitability, Performance Efficiency, and Maintainability** of NoteSphere.

The quantitative results will provide the basis for determining whether the developed system meets the intended software quality requirements and for identifying areas that may require further improvement.

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